

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13.	(a)	(i)		dipole shown on O—H bond using $\delta+$ and $\delta-$ (either water or diol) (1) lone pair of electrons shown on oxygen atom (either water or diol) (1) hydrogen bond shown between diol oxygen (lone pair) and hydrogen atom in water or between water oxygen (lone pair) and appropriate hydrogen atom in diol (1)		3		3		
		(ii)		award (1) for any of following • hydrogen bonding interferes with the formation of the ice lattice • hydrogen bonding stops the ice lattice from forming • disrupts the hydrogen bonding between water molecules • water molecules hydrogen bond more strongly with diol molecules than each other • water prefers to form hydrogen bonds with the diol than with itself neutral answer – hydrogen bonds form between molecules of the diol and water molecules			1	1		